

HANCHU ZHOU

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EDUCATION

University of California, Davis Ph.D. student in Electrical and Computer Engineering	Sep. 2022 - Present Davis, CA
Zhejiang University Bachelor's degree in Control Science (GPA: 3.89)	Sep. 2018 - Jun. 2022 Hangzhou, China

RESEARCH INTERESTS

World Model; Autonomous Driving; Robotics; LLM;

SKILLS

Programming:	Python, C, C++, MATLAB, HTML, Markdown, C#, JavaScript
Machine Learning:	JAX, TensorFlow, PyTorch, Scikit-learn, Pandas, OpenCV, LLM
Tools:	Open3D, YOLO, Matplotlib, Git

RESEARCH PROJECTS

CarDreamer: World-Model-Based Autonomous Driving Platform Apr 2024 - Jul. 2024
IEEE IoT-J: [CarDreamer](#) → ★180 stars

- Developed the first open-source platform for WM based autonomous driving in response to the lack of an easy-to-use WM based learning platform, with the goal of advancing research in this field.
- Designed various scenarios for training with great flexibility on modality, observability, and intention sharing, enabling users to explore diverse experiments with tailored settings. The trained DreamerV3 models show significant efficiency and safety in each scenario.

World-Model-Based Hierarchical Planning Mar 2024 - Jun. 2024
Under review in ICLR 2025

- Introduced HANSOME, a WM based hierarchical planning with semantic communication framework, aiming to address the challenges posed by partial observability and non-stationary in autonomous driving.
- Proposed AdaSMO, an entropy-controlled adaptive training method, to balance the training of higher-level and lower-level policies. The extensive experiments show that HANSOME outperforms SOTA WM-based methods in challenging driving tasks and enhances overall traffic safety and efficiency.

EI-Drive: Edge-Intelligent Autonomous Driving Platform Jun. 2023 - Aug. 2024
Under review in IEEE IoT-J: [EI-Drive](#)

- Developed a full-stack autonomous driving platform featuring edge communication and cooperative perception, enabling the exploration of V2X communication's impact on autonomous driving performance.
- Conducted extensive experiments based on various built-in scenarios and demonstrated that transmission latency and errors impair the performance of cooperative perception.

CommWorld: Communicative World Models for Distributed RL Feb. 2024 - May. 2024
Under review in ICLR 2025

- Propose CommWorld, the world model with lightweight information sharing, to mitigate the issue of insufficient information in distributed RL. Leveraging world models' generalization capability, CommWorld achieves better performance in the challenging planning experiments.

IterRob: The iteration of IL and RL in Robotics Training Dec. 2024 - Now
Ongoing research

- Fine-tuned the policies of robotics with Vision Language Model to specific tasks on ALOHA.
- Implemented the iteration of Imitation Learning and Reinforcement Learning to ensure the the learning efficiency and generalization capability of the models.